D71 — Parity-Driven Proton Leakage: A Monomer-on-Surface Mechanism Linking the Gravitational Bridge, the Closed Fraction and the Spin Budget

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Abstract

We report a mechanism-level result inside the coherence-network corpus (D-series): an odd-parity sea of $11017 = 23 \times 479$ relations on the proton surface forces exactly one conserved, mobile matching defect (a *monomer*), whose circulation, localised by an Euler condition at two degree-25 vertices on the bow of the spin-carrying valence core, yields an effective boundary count $B_{\text{eff}} = 528 + \frac{1}{2\sqrt{2}} + \frac{1}{56}$. This single mechanism, with all coefficients derived in closed form from the architecture integers, is simultaneously compatible with three independent observational targets: the proton-skin amplitude $\Delta r_{\text{val}} = 0.04795969 \pm 6.98 \times 10^{-8}$ controlling the $\alpha \leftrightarrow G$ bridge (residual 0.64σ), the closed fraction $f = 0.157126 \pm 9 \times 10^{-4}$ (candidate for Ω_b/Ω_m; residual 0.10σ), and the valence spin share $\Delta\Sigma = 0.30 \pm 0.05$ (EMC/COMPASS; residual 0.17σ). We distinguish carefully between theorem-level, derived-within-declared-family, and compatibility-level statements, and we show that this result does *not* yet lift the circularity of G: a continuous derivation C1–C4 $\to B \to \Delta r_{\text{val}} \to \varepsilon_G \to G_{\text{PDL}}$ remains to be written. A negative dossier (campaigns v3–v9) that eliminates six alternative branches is included as an integral part of the evidence.

1 Epistemic status and reading rules

Throughout, every statement carries one of three labels:

- **(T)** theorem-level: follows from the stated axioms C1–C4 and integer facts of the corpus, without free parameters;
- **(D)** derived within a family declared *a priori*, under an explicit Poisson null;
- **(H)** heuristic or compatibility-level: survives confrontation but is not uniquely selected.

The main claim of this note is at level **(D)**: one mechanism is simultaneously compatible with three independent numbers. Compatibility with three targets is not a derivation; Section 10 lists the exact falsifiers.

2 Corpus constants

3 The negative dossier (v3–v9)

Six natural branches were eliminated under declared families and Poisson nulls, which bounds the solution space from all sides:

- **v3:** naive irreversible closure freezes at 100%, not 15.7% — the closed fraction is not a raw one-shot process;

Quantity	Value	Role
Closure integer n_u	930	units of coherence
Sea vertices V_{sea}	918	dynamical sites
Sea relations E_{sea}	$11017 = 23 \times 479$	odd (parity engine)
Valence cores	$3 \times K_4$	12 vertices, 18 relations, static
Total relations	11035	sea + cores
Mean degree	24	sea coordination
Skin amplitude Δr_{val}	$0.04795969 \pm 6.98 \times 10^{-8}$	target T1 (bridge to G)
Closed fraction f	$0.157126 \pm 9 \times 10^{-4}$	target T2 (Ω_b/Ω_m candidate)
Valence spin $\Delta\Sigma$	0.30 ± 0.05	target T3 (EMC/COMPASS)

Table 1: Fixed inputs. All integers are corpus facts; the three targets are independent.

- **v5:** zero hit among 39 declared closed-form leak shapes (window $\pm 7 \times 10^{-8}$) — Δr_{val} is not an algebraic expression;
- **v6:** zero hit among 12 declared dynamical estimators of the closure trajectory (48 runs) — Δr_{val} is not a stochastic observable of a one-shot trajectory;
- **v7:** zero hit among 10 064 declared μ-deviation forms; the available μ discrepancy carries $\sim 0.00\%$ of the required leak (a factor $\sim 5 \times 10^4$ too small, wrong geometry). The bridge target itself shifts by 12 tolerances under the $\mu_{\text{pred}}/\mu_{\text{exp}}$ convention;
- **v8:** 52 declared single-level stationary mean-field fixed points $p^* = U/(U + L)$ falsified; required suppression measured: 14.24 bits;
- **v9:** 1080 declared hierarchical period-3 $\times$ spin configurations; zero triple coincidence; maximum per-level suppression 7.16 bits at level 1 versus 0.79 bits/level needed over 18 levels.

Convergent conclusion at level **(H)**: the leak must be a stationary, analytically derivable property of a *fixed-parity* ensemble — the grand-canonical homogeneous treatment used in v8/v9 was the wrong ensemble.

4 Parity theorem and the conserved monomer (v10)

Theorem (T, within the matching model). Since $E_{\text{sea}} = 11017 = 23 \times 479$ is odd, every matching of the sea leaves at least one unmatched relation (monomer). Creating a second defect costs $\sim 2^{1-k}$ broken locks per locked neighbour (exact law, corpus theorem U2), hence the equilibrium state carries *exactly one* conserved, mobile monomer.
Consequences:

- the unlocking rule U — postulated in v8/v9 — becomes *derived*: the defect cannot be eliminated, only displaced (level T given the model);
- the stationary distribution of the monomer is analytically solvable (exact Markov-chain fixed point; validated to 1.9×10^{-3});
- inverting the skin target yields $B_{\text{req}} = 528.354$ effective boundary relations — *non-integer*, which falsified the uniform-boundary family at ppm level and localised the obstruction to a fractional boundary correction of 0.354.

5 Surface ontology and the wake census (v12)

The proton is treated as a *surface only* (soap-bubble / black-hole-horizon ontology): no bulk; all relations live on the membrane. The sea is a sparse dynamical graph; the three K_4 valence cores

are complete graphs moving through it like bows through water, carving *wakes*. Degree audit of the sea: $\sum \deg = 22034 = 918 \times 24 + 2$ — hence **exactly two vertices of degree 25** carry the parity excess (level T given the model).
Wake boundary count (level D): orthogonal + diagonal front = 540; minus one core's diagonal interface 12:

$$B = 540 - 12 = 528 = n_u\,(n_u - 2) = 24 \times 22 = 12 \times 44.$$

The residual 0.354 is then hit by the fractional correction $B = 528 + \frac{1}{2\sqrt{2}} = 528.35355339$ versus target 528.35404759: deviation 0.000447, i.e. 0.58σ inside the declared family (2496 forms, Poisson $\lambda = 0.064$, $P(\geq 1) = 6.2\%$). Signature of a *diagonal cut*: a bow crossing a mesh diagonally contributes half-edges weighted by $\sin(\pi/4)$.

6 Euler localisation of the defect (v13)

(D) An alternating defect trail on the wake boundary is Eulerian iff the number of odd-degree vertices of the front is 0 or 2. The diagonal interface of a core is 12 (even); hence the degree excess must arrive *in pairs*: two degree-25 vertices, not one of degree 26 (which would break alternation). Scenario exclusion fixes both loci *on the bow of the reference (spin-carrying) core*. Per locus, the diagonal-stub correction is

$$\frac{\sqrt{2}}{2} \times \frac{1}{2} \times \frac{1}{2} = \frac{\sqrt{2}}{8}, \qquad 2 \text{ loci} \Rightarrow \frac{\sqrt{2}}{4} = \frac{1}{2\sqrt{2}},$$

and the naive duty cycle $B_{\text{duty}} = 528.372$ is reconciled with the inverted target by the stationary correction $\delta = 1/(n_u + 32) = 1/56$: $528.372 - 1/56 = 528.354$.

7 Triple-target confrontation (v14)

The frozen mechanism was confronted with all three targets simultaneously:

- **T1 (pass):** $B_{\text{pred}} = 528 + \frac{1}{2\sqrt{2}} + \frac{1}{56}$, giving $\Delta r_{\text{val}}^{\text{pred}} = 0.04795965$ vs 0.04795969 — residual $4.5 \times 10^{-8} = 0.64\sigma$;
- **T2 (pass):** among 29 declared forms, 7 hits; selected form $f = 12^2/(V_{\text{sea}} - 1) = 144/917 = 0.15703381$ vs 0.157126 — residual 0.10σ. Note $917 = V_{\text{sea}} - 1$: the same parity defect governs skin and closed fraction;
- **T3 (pass, weakly discriminating):** candidates within the wide EMC window.

Survival of three targets is *compatibility*, not derivation (level H at this stage).

8 Closed-form derivations (v15–v16): agenda 5/5

1. **Bow degree (closed):** $d_{\text{bow}} = n_u/3 = 8$, the per-core closure integer.
2. **Duty correction (closed):** $\delta = 1/\binom{8}{3} = 1/56$ — each bow locking commits one tetrahedral face among $\binom{8}{3} = 56$ equiprobable configurations; unique argmin (0.124σ) in the declared $\binom{d}{k}$ family, $d \in \{6, 8, 12, 24\}$, $k \in \{2, 3, 4\}$.
3. **Diagonal cut (closed):** $\sqrt{2}/8$ per Euler locus is the *unique* element of the declared $g \times o \times p$ family passing at $< 1\sigma$.
4. **Closed fraction (closed):** $f = 144/917$ selected over $2/(3\varphi^3)$ by the structural chain criterion (φ lies outside the wake chain; the denominator must include the monomer, $917 = V_{\text{sea}} - 1$). The v3 fatality of $2/(3\varphi^3)$ is thereby *confirmed and reconciled*: distinct readings, distinct windows; Poisson null $P = 9.3\%$ for its v14-window reappearance as coincidence.

5. **Valence spin (closed):** from the spin–surface map,

$$\Delta\Sigma = \underbrace{\tfrac{1}{3}}_{\text{accessible phase (D69)}} \times \underbrace{\left(1-\tfrac{1}{8}\right)}_{\text{bow relations unabsorbed by the wake}} = \frac{7}{24} = 0.29167,$$

vs 0.30 ± 0.05: 0.167σ. The competing exact form $3/10$ and $1/(2\varphi)$ are excluded as outside the structural chain.

Cross-coherence (the strongest internal signature): the single integer pair $(3, 8)$ governs *both* T1 ($\delta = 1/\binom{8}{3}$) and T3 ($\Delta\Sigma = \frac{7}{24} = \frac{1}{3}(1-\frac{1}{8})$). Two independent targets, one pair of integers. The implied spin budget reads: valence $7/24$, sea + orbital $17/24 \approx 0.71$ — consistent with an almost entirely dynamical energy budget (QCD).

9 Results and residuals

Target	Mechanism prediction	Observation	Residual
T1 Δr_{val}	0.04795965	$0.04795969 \pm 6.98 \times 10^{-8}$	0.64σ
T2 f	$144/917 = 0.15703381$	$0.157126 \pm 9 \times 10^{-4}$	0.10σ
T3 $\Delta\Sigma$	$7/24 = 0.29167$	0.30 ± 0.05	0.17σ

Table 2: Triple-target audit after v16.

10 Limitations and falsifiers

1. The fractional T1 hit is at $\sim 1.7\sigma$ (Poisson $\lambda = 0.064$, $P(\geq 1) = 6.2\%$): a serious indication, not a discovery.
2. The 7 T2 hits are correlated (single counting cluster); the nominal 9×10^{-8} overstates significance. Selection was by structural criterion, not by fit.
3. T3's EMC window (± 0.05) is wide: T3 confirms without strongly discriminating.
4. The matching-model reduction (sea as dimer/monomer problem under the 2^{1-k} law) is itself level (D), not (T); the graph spectral dimension ≈ 2 (the *surface* claim) remains to be established, not assumed.
5. The "external" agent maintaining total spin 1/2 (the closure wave) must be identified: condition at measurement, or global 930-closure as integral constraint.
6. **G remains circular.** What is achieved is a fully specified mechanism whose every coefficient is derived in closed form; what is missing is the continuous demonstration C1–C4 $\to B \to \Delta r_{\text{val}} \to \varepsilon_G \to G_{\text{PDL}}$ that never consults the measured α.

Engraved falsifier (pre-registered): any admissible mechanism must reproduce *simultaneously* $\Delta r_{\text{val}} = 0.04795969 \pm 6.98 \times 10^{-8}$ and $f = 0.157126 \pm 9 \times 10^{-4}$ from a single law; a law predicting only one of the two is dead.

11 Reproducibility

All campaigns are stdlib-only, Colab-ready Python scripts with declared families printed before resolution and embedded pre-registration blocks dated 2026-10-29: v5 (leak closed forms), v6 (dynamical estimators), v7 (μ deviations), v8 (single-level mean field), v9 (hierarchical period-3 + spin), v10 (parity monomer, exact stationary law), v12 (wake census), v13 (Euler localisation), v14 (triple target), v15 (closed forms I), v16 (closed forms II). Archive: https://cloud.giotto.ai:9443/default/orchestrator-uploads/cli-uploads/ (paths per campaign: `pdl-leak-v5` ... `pdl-leak-v16`, `pdl-wake-v11`, `pdl-wake-v13`).

12 Conclusion

A single parity-driven monomer mechanism on the proton surface — odd sea $\Rightarrow$ one conserved defect, Euler-localised at two degree-25 loci on the spin core's bow, diagonal wake cuts contributing $1/(2\sqrt{2})$, duty correction $1/\binom{8}{3}$ — is simultaneously compatible with the proton skin feeding the $\alpha \leftrightarrow G$ bridge, the cosmological closed fraction, and the valence spin share, with every coefficient derived from architecture integers within declared families. The next and decisive step is the continuous derivation of G_{PDL} from C1–C4 without recourse to measured α, or a clean falsification of the mechanism on the engraved double window.

References

[1] EMC Collaboration, *Phys. Lett. B* **206** (1988) 364; COMPASS and HERMES spin measurements (valence spin share $\Delta\Sigma \approx 0.30 \pm 0.05$).

[2] R. L. Jaffe, A. Manohar, *Nucl. Phys. B* **337** (1990) 509 (spin decomposition).

[3] CODATA recommended values of the fundamental physical constants (bridge target G, α).

[4] C. Laubscher, coherence-network corpus D21–D70 (this archive, 2026): closure integer 930, sea integers 918/11017, period-3 cycle (D69), μ operator chain (D28–D30), U-series theorems (U1v2 minimal decomposition $310\varphi + 310 + 310/\varphi^2$; U2 lock law 2^{1-k}).